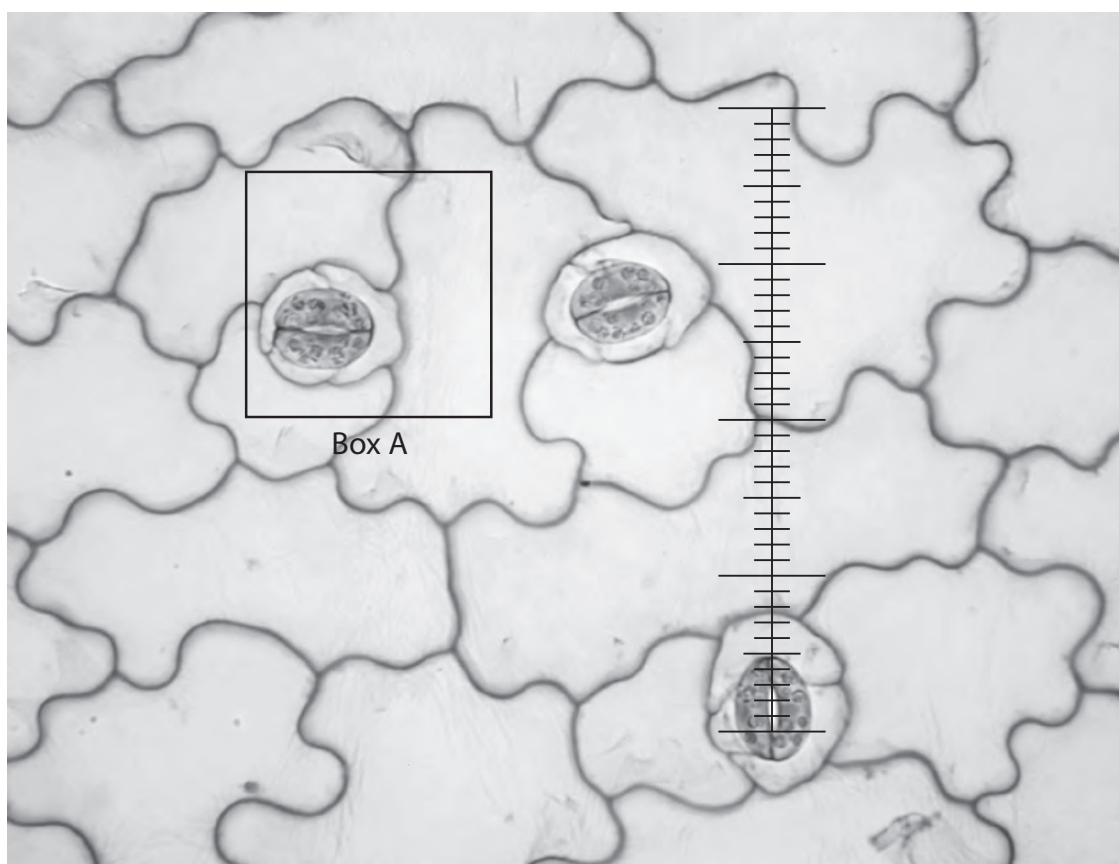


- 3 This photograph, taken through a light microscope, shows three of the stomata on the underside of a leaf.



(Source: John Adds)

- (a) (i) Draw the cells within Box A on the photograph.

(3)

DO NOT WRITE IN THIS AREA

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(ii) An eyepiece graticule is shown over one of the pores.

Each of the smallest units on the graticule is $3 \times 10^{-6} \text{ m}$.

Calculate the length of this pore in micrometres (μm).

(2)

Answer μm

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DO NOT WRITE IN THIS AREA



P 6 2 4 6 0 A 0 1 5 2 0

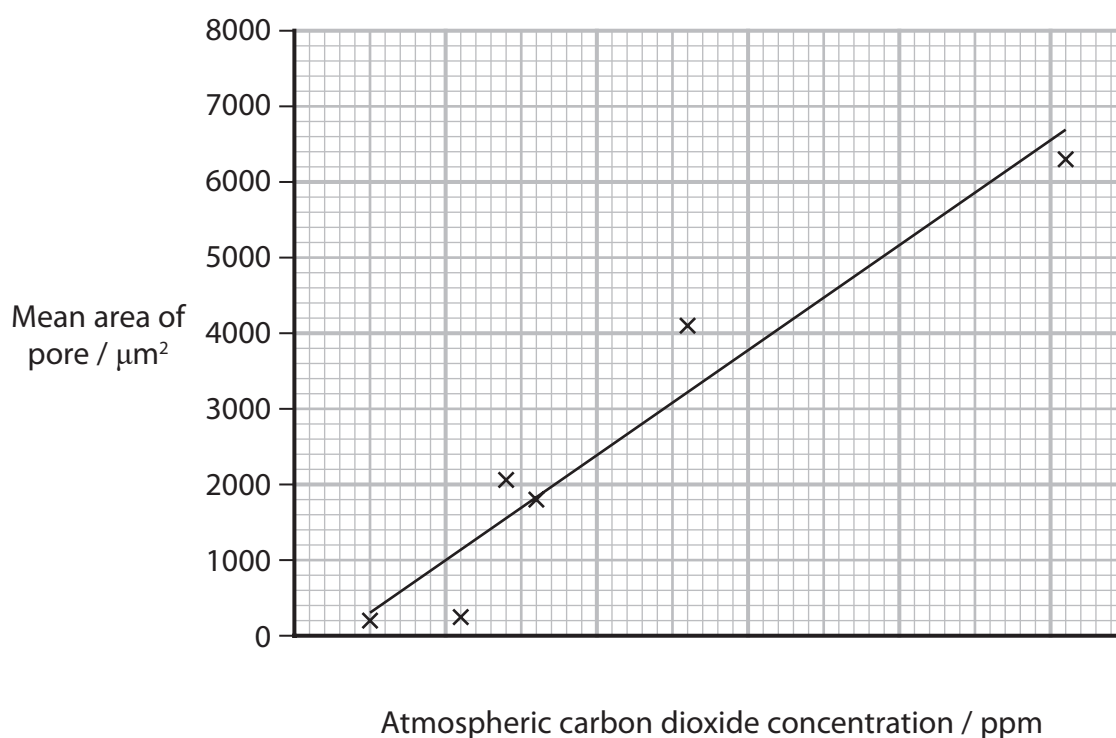
- (b) The area of a sample of pores in fossil leaves was investigated and the means calculated.

These means were correlated with estimates of atmospheric carbon dioxide concentration at the time the fossil was formed.

The results are shown in the table.

Atmospheric carbon dioxide concentration / ppm	Mean area of pore / μm^2	Standard deviation
250	200	1
550	240	2
700	2050	150
800	1800	70
1300	4100	160
2550	6300	1600

The data in the table were plotted on a graph.



(i) Add the scale for carbon dioxide concentration to the graph. (1)

(ii) Draw the standard deviation for the mean area of the pores $4100\mu\text{m}^2$ and $6300\mu\text{m}^2$ on the graph. (1)

(iii) It was concluded that the mean areas of these pores were significantly different.
Justify this conclusion. (2)

(iv) Predict the mean area of pores at a time in the Earth's history when the carbon dioxide concentration was 6500 ppm.
Use the equation $y = mx + c$, where $m = 2.8$ and $c = -397.5$ (2)

Answer μm^2

(Total for Question 3 = 11 marks)

TOTAL FOR PAPER = 50 MARKS

